Economic Factors and Happiness

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1. Motivation

The question that started this project was simple: does money actually make people happy? Politicians talk about GDP growth as if it is the ultimate measure of societal progress, but I always found this a bit suspicious. Does it matter how fast an economy is growing, or is it more about how wealthy a country already is and does wealth even matter more than social factors like community and support?

My going-in hypothesis was that economic variables, particularly income level and unemployment, would be the strongest predictors of happiness, with GDP growth rate also playing a meaningful role. I expected the data to confirm the intuition that richer, faster-growing economies produce happier citizens.

As it turned out, the data challenged that hypothesis in an interesting way. My main research questions were:

- Can macroeconomic indicators (income level, unemployment, inflation, GDP growth) explain differences in national happiness?
- Which factors (economic or social) are most predictive of happiness?
- Does GDP growth rate matter, or is it the existing level of wealth that counts?

2. Data Source

World Happiness Report (2015–2024)

I downloaded the World Happiness Report data from Kaggle. It covers roughly 150 countries per year and includes not just happiness scores but also factors like social support, life expectancy, freedom, generosity, and perceived corruption. I chose this dataset because it is well-known, widely cited in academic research, and already cleaned to a reasonable degree, which made it a reliable foundation to build on.

World Bank Open Data (2015–2024)

To add a purely economic dimension, I pulled three indicators directly from the World Bank's open data portal:

- GDP per capita growth (annual %)
- Inflation, GDP deflator (annual %)
- Unemployment, total (% of labor force)

I chose these three specifically because they represent different dimensions of economic conditions: GDP growth captures momentum, inflation captures price stability, and unemployment captures labor market health. These came in wide format (one column per year), so I had to reshape them into long format before merging.

Merging & Cleaning

I merged the two datasets on Country and Year. The raw merged dataset had 1,502 rows. The World Bank indicators had missing values, 217 missing for GDP Growth and Inflation, 224 for Unemployment, mostly from small or conflict-affected countries where data collection is difficult.

I decided to drop rows where any of these three indicators were missing, reducing the dataset to 1,272 rows. I considered imputation, but given that the missing countries tend to be structurally different (smaller economies, conflict zones), imputing values for them felt like it would introduce more noise than it would remove.

One thing I noticed early on: the happiness dataset uses GDP per capita as a log-transformed index, not a raw dollar amount. This is already a measure of wealth level, not growth, which became relevant later when comparing it to the *World Bank's GDP growth rate variable.*

3. Data Analysis

3.1 Exploratory Data Analysis

Before building any models, I wanted to understand what the data actually looked like and which variables had meaningful relationships with happiness. I started with a correlation matrix.

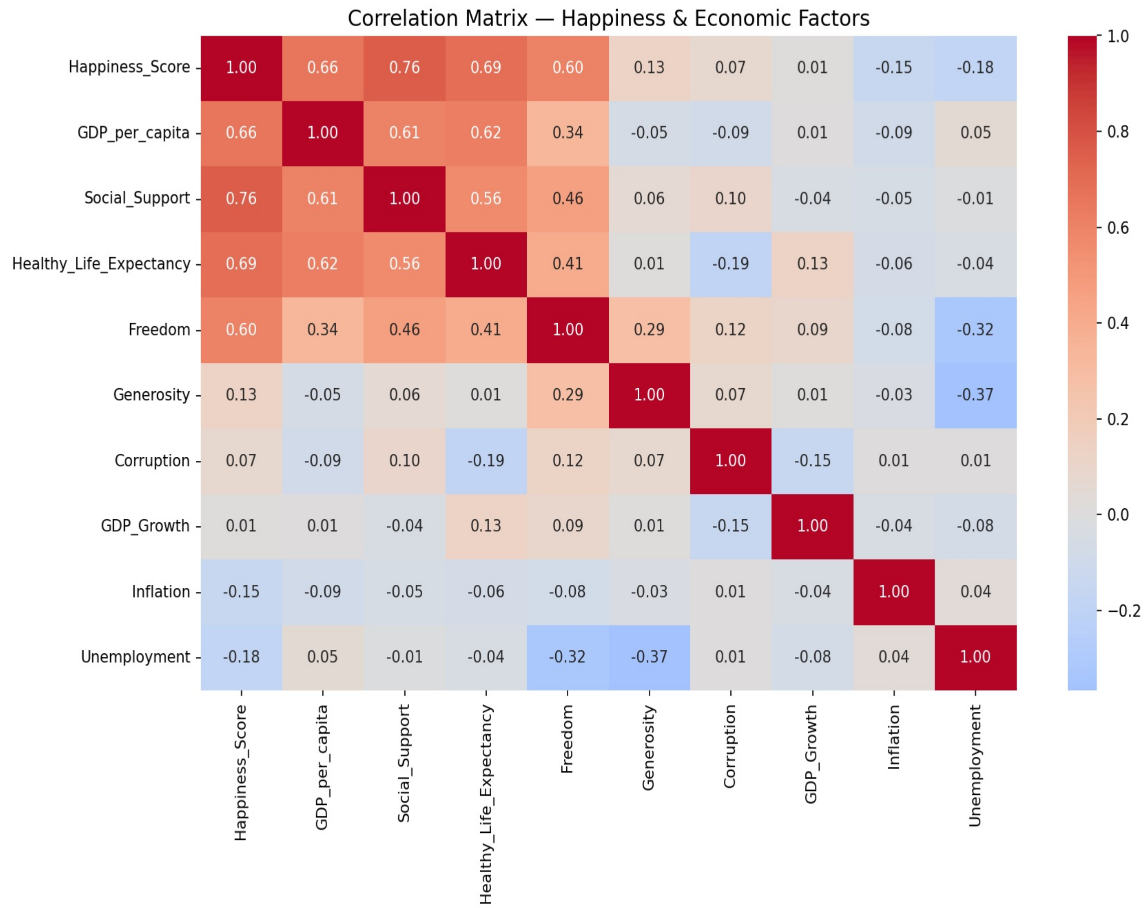

Figure 1: Correlation Matrix — Happiness & Economic Factors.

Social Support showed the strongest correlation with happiness (0.76), followed by GDP per capita (0.66). Notably, GDP Growth had virtually no correlation (0.01), suggesting that economic momentum alone does not drive well-being.

The correlation results immediately challenged my initial hypothesis. Social Support had the strongest correlation with happiness (0.76), stronger than GDP per capita (0.66). This was surprising, I expected income to dominate. But it makes sense: even in wealthy countries, people who feel isolated or unsupported tend to report lower happiness.

What was perhaps even more striking was GDP Growth: it had a correlation of essentially zero (0.01) with happiness. This already suggested, before any modeling, that the rate of economic expansion has little to do with how happy a population is.

Variable	Correlation with Happiness
Social Support	0.76 (strongest)
Healthy Life Expectancy	0.69
GDP per capita	0.66
Freedom	0.60
Generosity	0.13

Variable	Correlation with Happiness
Corruption	0.07
Inflation	-0.15
Unemployment	-0.18
GDP Growth	0.01

Figure 2: GDP per Capita vs. Happiness Score by Region

Countries with higher GDP per capita tend to report higher happiness scores. However, the relationship is nonlinear, the happiness gains from increased wealth are much larger at lower income levels, suggesting diminishing returns. Regional clustering is also visible: Western European countries appear consistently at the top, while Sub-Saharan African countries cluster at the lower left.

The scatter plot reveals an important nuance: the relationship between GDP and happiness is not linear. Countries at the lower end of the income scale see much larger happiness gains per unit of GDP than already-wealthy countries. This nonlinearity became relevant when comparing linear vs. tree-based models later.

One regional pattern worth noting: Latin America ranks higher than its income levels would predict. This is consistent with the so-called 'Latin American happiness paradox', suggesting that cultural factors like family ties and community cohesion may compensate for economic disadvantage.

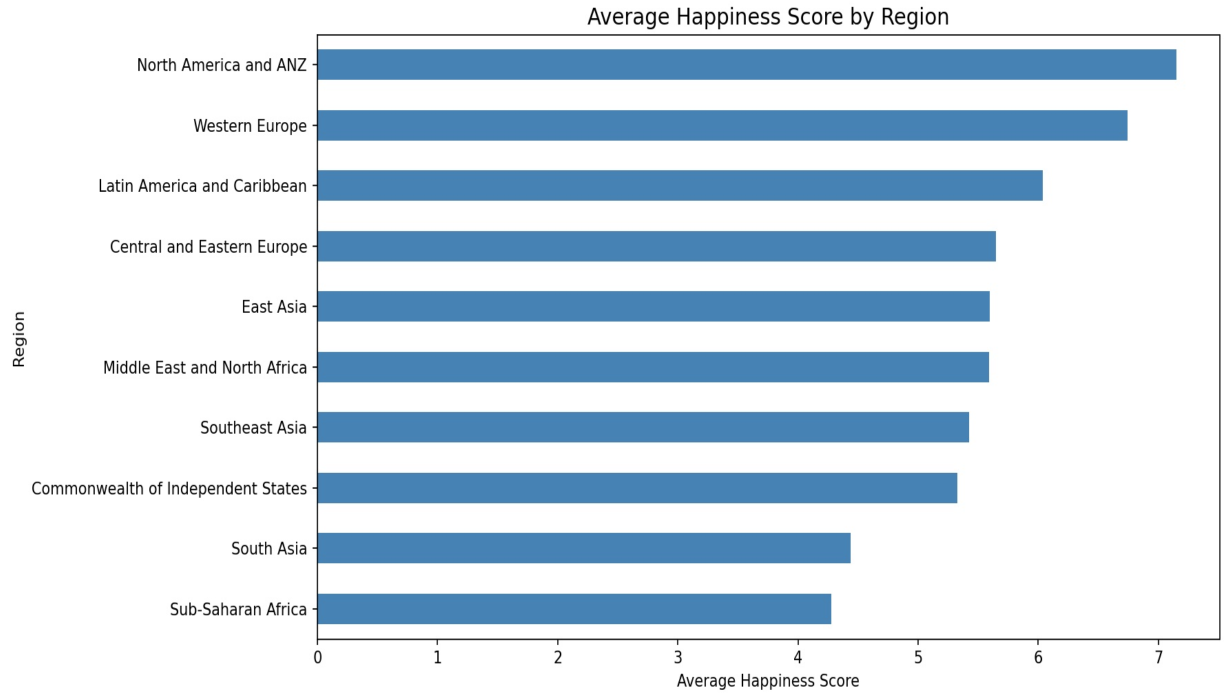

Figure 3: Average Happiness Score by Region (2015–2024)

North America & ANZ and Western Europe consistently rank as the happiest regions, while Sub-Saharan Africa and South Asia have the lowest average scores. Interestingly, Latin America ranks higher than expected given its income levels, which suggests that factors beyond economics, such as social bonds and cultural attitudes, play a significant role in happiness.

3.2 Hypothesis Testing

To check whether the correlations I observed were statistically meaningful, I ran Pearson correlation tests for each of the four economic variables against Happiness Score:

Variable	Correlation	p-value	Result
GDP per capita	0.657	< 0.0001	Significant
GDP Growth	0.012	0.6727	Not Significant
Inflation	-0.149	< 0.0001	Significant
Unemployment	-0.180	< 0.0001	Significant

The key result: GDP growth rate is not significantly associated with happiness ($p = 0.67$), while income level, unemployment, and inflation all are. A country growing at 5% per year is not necessarily happier than one growing at 1%, what matters is how wealthy the country already is, not how fast it is expanding.

This finding changed how I think about economic policy discussions. Much political debate focuses on quarterly growth figures, but the data suggests that for happiness, what you already have matters far more than how fast you are gaining more of it. A poor country growing fast is still a poor country.

3.3 Machine Learning Models

For the modeling phase, I used all 9 features and split the data 80/20 for train and test. Ridge and Lasso were applied with feature scaling to ensure all variables are on the same scale before modeling.

I chose to compare five models: Linear Regression as an interpretable baseline, Ridge and Lasso as regularized linear alternatives (useful given the multicollinearity in the data), and Random Forest and Gradient Boosting as ensemble tree methods. I intentionally excluded models like SVM and KNN because they are harder to interpret and their advantages typically show at larger dataset sizes, with 1,272 observations and 9 features, tree ensembles are well-suited and provide built-in feature importance measures.

Cross-Validation Results (5-Fold)

Model	CV R^2	Std
Random Forest	0.830	±0.020
Gradient Boosting	0.820	±0.019
Linear Regression	0.740	±0.041
Ridge (a=1)	0.741	±0.041
Lasso (a=0.1)	0.730	±0.020

Test Set Performance

Model	R^2	MAE	RMSE
Gradient Boosting (Tuned)	0.823	0.353	0.477
Random Forest	0.821	0.348	0.476
Linear Regression	0.742	0.451	0.571
Ridge (a=1)	0.742	0.451	0.571
Lasso (a=0.1)	0.694	0.491	0.622

The gap between linear and tree-based models confirms there are meaningful nonlinear relationships in the data. The GDP–happiness scatter plot already hinted at this: diminishing returns to wealth cannot be captured by a straight line.

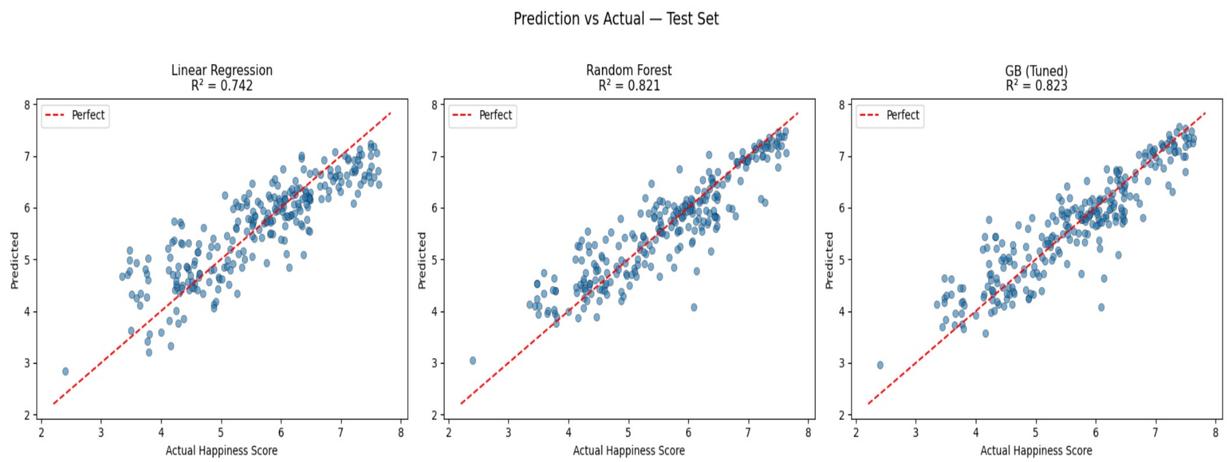

Figure 4: Prediction vs. Actual Happiness Scores

Gradient Boosting and Random Forest predictions align much more closely with the actual values compared to Linear Regression, particularly at the lower and upper ends of the happiness scale.

Hyperparameter Tuning

To improve Gradient Boosting performance, I tuned the model using grid search with 5-fold cross-validation. I tested different combinations of tree depth, number of trees, and learning rate. The best combination turned out to be learning rate=0.1, max depth=4, and n estimators=200, which gave a cross-validation R^2 of 0.832 and a test R^2 of 0.823.

Learning Curves & Overfitting

The initial learning curves showed a large gap between train R^2 (~0.98) and validation R^2 (~0.65–0.70), a clear sign of overfitting. I applied regularization by adding min samples leaf and subsample constraints, which significantly reduced the gap and improved generalization.

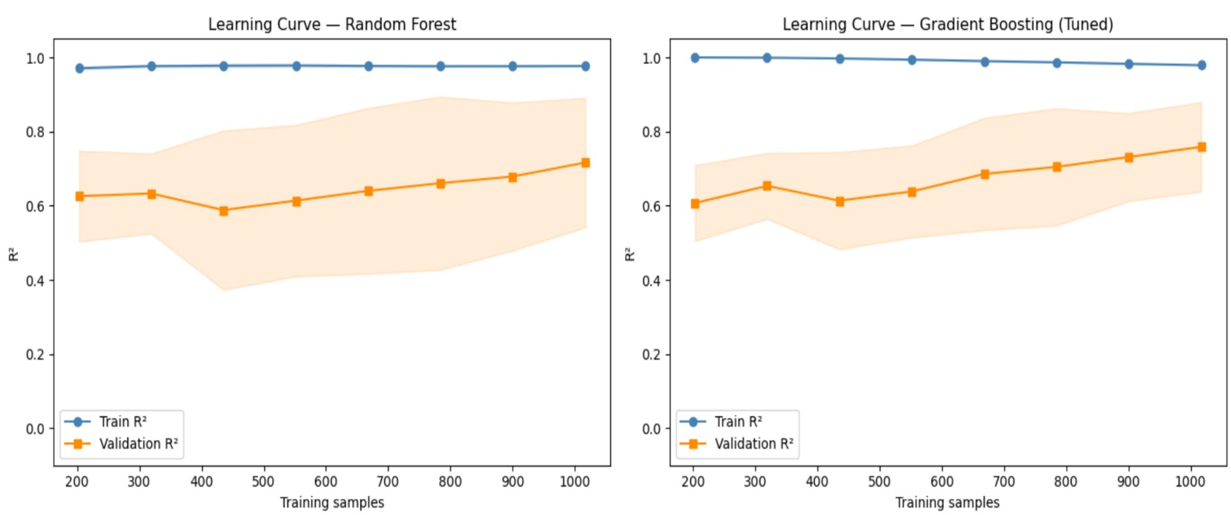

Figure 5: Learning Curves (Before Regularization)

The training score stays near 1.0 while the validation score is much lower, which indicates the model is overfitting, it performs well on training data but struggles to generalize to new data.

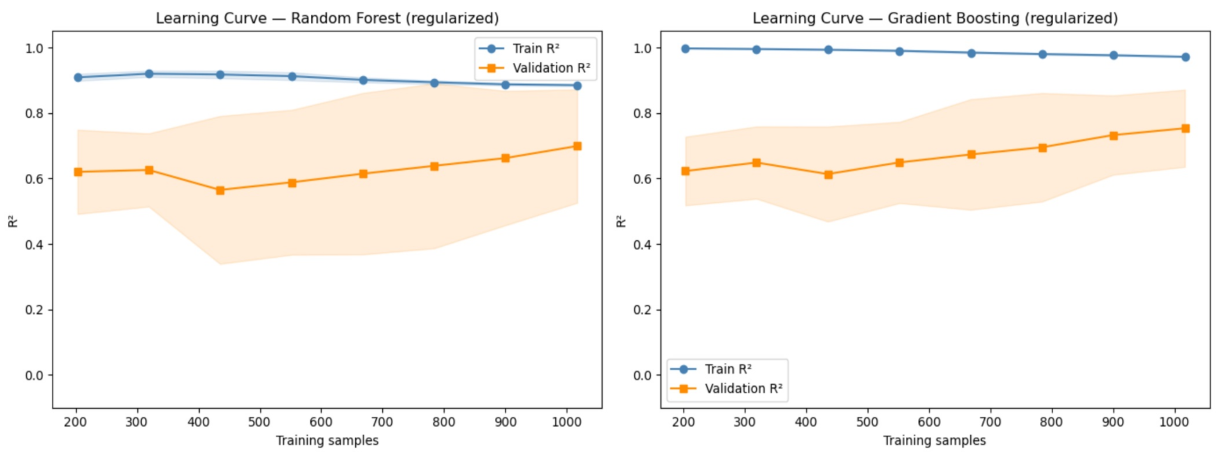

Figure 6: Learning Curves (After Regularization)

After applying regularization, the gap between training and validation scores narrows significantly. The regularized Gradient Boosting model achieved the best overall performance with R^2=0.841 and MAE=0.327.

Feature Importance

Both Random Forest and Gradient Boosting ranked Social Support as the dominant feature (~0.50–0.55 importance), with GDP per capita a distant second (~0.15). The three World Bank economic indicators I added, GDP Growth, Inflation, Unemployment, all had very low importance scores. The story of happiness, at the national level, is overwhelmingly about social and structural well-being, not short-term economic fluctuations.

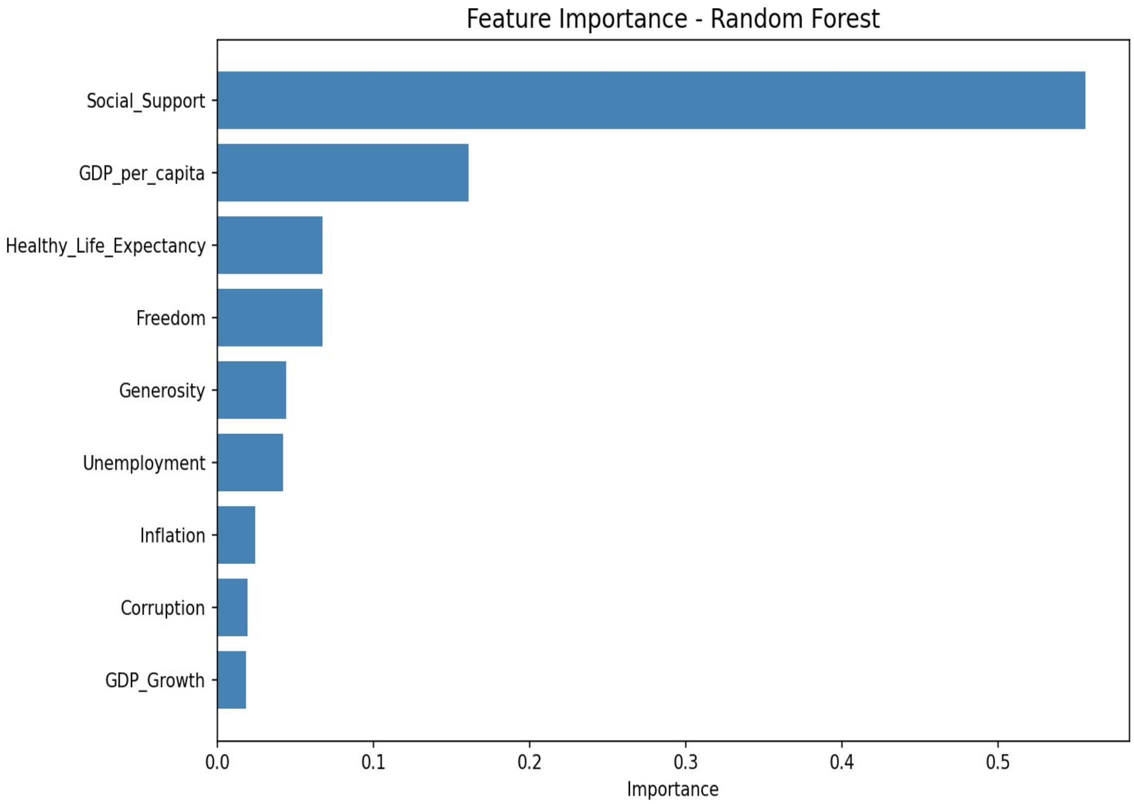

Figure 7: Feature Importance

3.4 Lasso Feature Selection

Lasso regression is particularly interesting because it forces feature selection by shrinking irrelevant coefficients to exactly zero. I ran Lasso with cross-validation to find the best regularization strength and observe which features survive.

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Lasso Coefficients:
Unemployment             -0.084715
Inflation                -0.075307
GDP_Growth               -0.000000
Generosity               -0.000000
Corruption                0.056035
Freedom                   0.216411
GDP_per_capita            0.218924
Healthy_Life_Expectancy   0.285134
Social_Support            0.450322
dtype: float64

Features eliminated (coef=0):
['GDP_Growth', 'Generosity']

Features kept:
['GDP_per_capita', 'Inflation', 'Unemployment', 'Social_Support', 'Healthy_Life_Expectancy', 'Freedom', 'Corruption']
```

Figure 8: LassoCV results

Lasso eliminated GDP Growth and Generosity completely, setting their coefficients to zero. This means that once other features are controlled for, these two variables add no independent explanatory power. GDP Growth being eliminated is consistent with the hypothesis test result, growth rate simply does not predict happiness. The remaining seven features all survived, with

Social Support having the strongest positive effect and Unemployment and Inflation having negative effects.

Feature	Coefficient	Status
Social Support	+0.450	Kept
Healthy Life Expectancy	+0.285	Kept
GDP per capita	+0.219	Kept
Freedom	+0.216	Kept
Corruption	+0.056	Kept
Unemployment	-0.085	Kept
Inflation	-0.075	Kept
GDP Growth	0.000	ELIMINATED
Generosity	0.000	ELIMINATED

GDP Growth eliminated: Growth rate simply does not predict happiness once you control for everything else. A country can grow quickly yet remain poor and unhappy, while a rich but stagnant economy maintains high well-being.

Generosity eliminated: Its raw correlation with happiness was already weak (r=0.13). Once other features are controlled for, Generosity adds no independent explanatory power, generous behavior may be more a symptom of happiness than a cause of it.

Unemployment and Inflation survived: Both retained non-zero negative coefficients, confirming that macroeconomic instability suppresses happiness even after controlling for wealth level.

The Lasso results gave me more confidence in the overall story. Without any prior assumption, it independently arrived at the same conclusion as the hypothesis tests: GDP growth is irrelevant, but income level, social support, health, freedom, and basic economic stability all matter. Two methods converging on the same answer is reassuring.

4. Findings

Stepping back from the individual analyses, the data tells a consistent and somewhat counterintuitive story: economic growth, the metric most obsessed over in policy and media, has essentially no relationship with national happiness. What matters instead is a combination of structural wealth (GDP level), social cohesion, physical well-being, and basic economic stability. Here are the five key takeaways:

1. Social support is the strongest predictor of national happiness, stronger than any economic variable. It had the highest correlation (0.76), the highest Lasso coefficient (0.450), and the highest feature importance in both tree models. Economic factors alone are insufficient to explain happiness.
2. Income level matters; growth rate does not. GDP per capita is strongly and significantly related to happiness (r=0.66, p<0.0001), while GDP growth rate is not (p=0.67) and was

completely eliminated by Lasso. Policymakers focused exclusively on growth may be optimizing the wrong metric for societal well-being.

3. Unemployment and inflation suppress happiness, confirmed by both hypothesis tests and Lasso coefficients, even after controlling for income level and life quality. Economic instability has a real cost to well-being beyond just reducing income.
4. Nonlinear models substantially outperform linear ones. The gap between ensemble and linear models reflects meaningful nonlinear relationships. The best model explained about 84% of variance in happiness scores.
5. Lasso independently confirmed the redundancy of GDP Growth and Generosity, providing a parsimonious picture: social support, health, income level, freedom, and macroeconomic stability are what determine national happiness.

5. Limitations & Future Work

Limitations

- Causality: Everything here is correlational. I cannot prove that wealth causes happiness, it could run the other direction, or both could be driven by a third factor like institutional quality. Wealthier countries also tend to have better statistical offices, which may affect data quality.
- Missing data and selection bias: Dropping 230 rows (~15%) to handle missing World Bank values introduced selection bias. Countries with missing data tend to be smaller or conflict-affected, precisely those where we might expect unusual happiness scores.
- Happiness data is subjective and survey-based: The World Happiness Report relies on self-reported life satisfaction scores. Cultural differences in how people express happiness, not just actual differences in well-being, may affect cross-national comparisons.
- 2020 COVID outlier effect: The 2020–2021 period was highly unusual economically and socially. Happiness scores, unemployment rates, and GDP figures from this period may be outliers that distort patterns from more normal years. The analysis did not treat this period separately.
- Multicollinearity: Many features are highly correlated (e.g., Social Support and GDP per capita: r=0.61), making individual coefficient interpretation unreliable for linear models.
- Panel structure ignored: I treated observations as independent, but a fixed-effects panel regression would better account for country-specific baselines across the 10-year period.

Future Work

- Apply fixed-effects panel regression to properly exploit the time-series structure and control for country-specific factors.
- Add inequality measures (Gini coefficient) and political freedom indices, richer countries with high inequality may have lower happiness than their GDP suggests.

- Separate the 2020–2022 COVID period and model it independently to see whether the drivers of happiness shifted during the pandemic.
- Cluster countries into groups by income level or region and build separate models per cluster.

AI Tools Usage

Stage	Usage
Data cleaning strategy	Discussed whether to drop or impute missing World Bank rows
ML model selection	Guided choice of models and cross-validation approach
Lasso analysis	Suggested LassoCV code to address learning assistant feedback on feature selection
Report writing	Helped structure and draft sections; all content reviewed and finalized by the author
Debugging	Helped identify and fix code errors during the analysis

All code was written and executed by the author. AI was used for guidance and drafting support, not for generating final outputs autonomously.